

CS4800: Designing Sustainable ICT Systems

Lecture 5.1: Sustainable Cloud Computing, Part 1

September 25, 2026

Lecturer: Przemysław Pawełczak

Authors: Alex Nedelcu, Cristian Cuțitei, Przemysław Pawełczak

CS4800-EWI@tudelft.nl

What will you learn in this lecture?

Lecture 5.1: Sustainable Cloud Computing, Part 1

- Data center architecture: computing, cooling, power, networking
- Environmental footprint of cloud computing, measurement methods

What is a data center?

Definitions: data centers

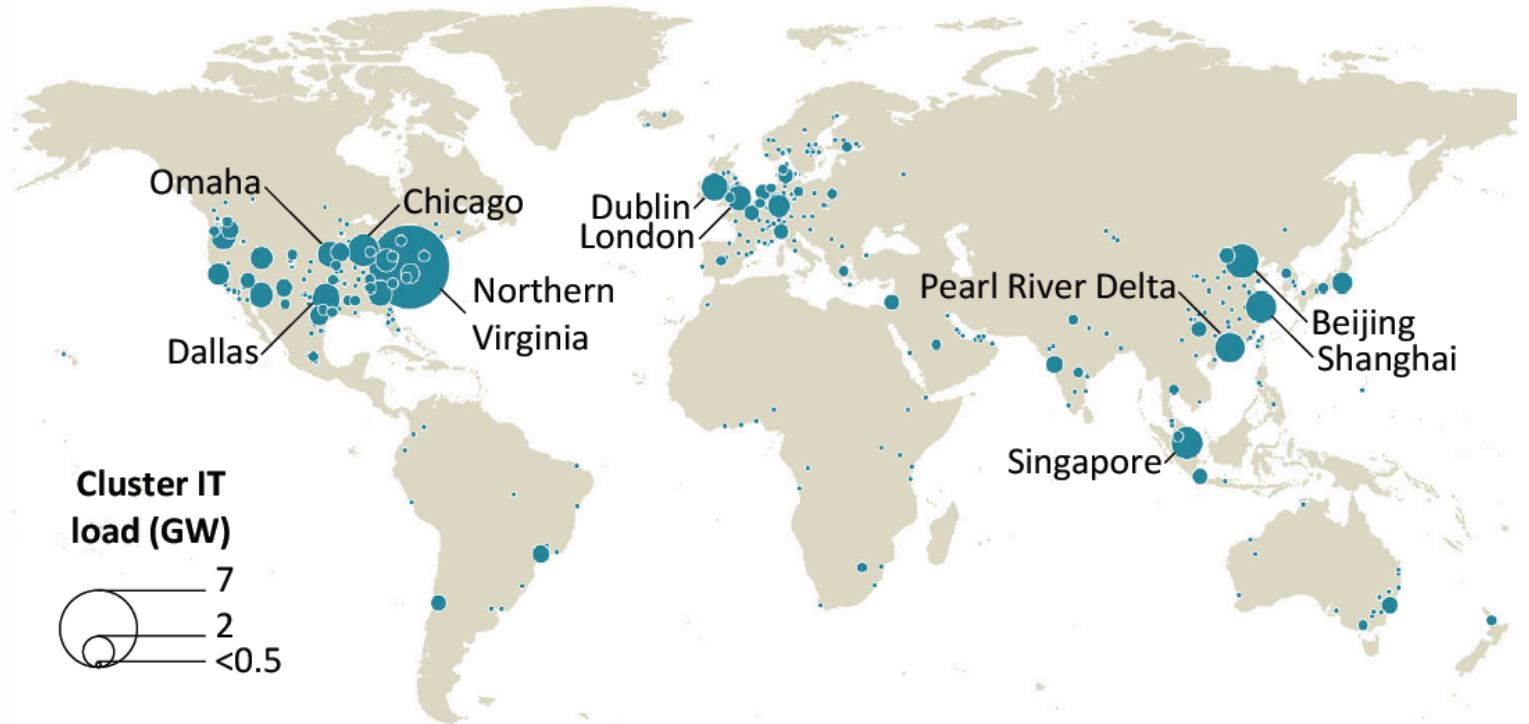
- Data center: space that contains ICT equipment (from closet-sized server to >10,000 sqm)
- Rough [classification](#) by usage/[power load](#):
 - Internal/on-prem/enterprise: used by a single company, [<1MW](#)
 - Wholesale/colocated: customers rent portions, [~1-100MW](#)
 - Hyperscale: built by large corporations for their own operations, [>100MW](#)
 - Give reference household consumption
- Utilization rate: percentage of time that computers are in operation (<50% for [enterprise loads](#), >80% for AI training)



[A lot of computers in a big warehouse](#)

Data centers: locations

- High spatial concentration: consume [20% of electricity supply](#) in Ireland; 25% in Virginia, US
- Majority of capacity (incumbent and under construction) in the US
- Large clusters around North-Western Europe: Frankfurt, London, Amsterdam, Dublin
- Growing clusters around Singapore, Beijing, Shanghai, Middle East

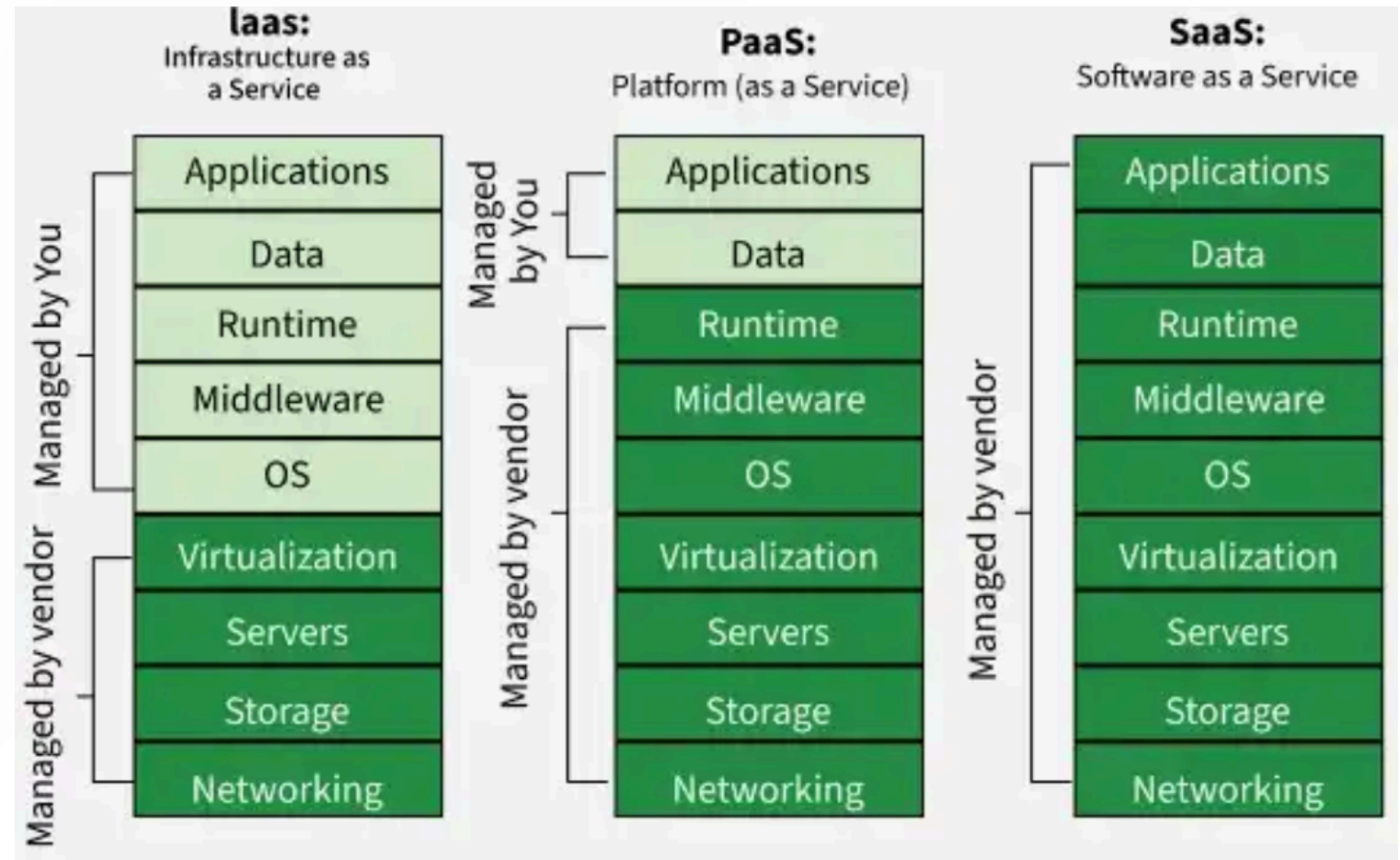


[Global concentration of data centers](#)

Data centers: activities

High-level [use-cases](#):

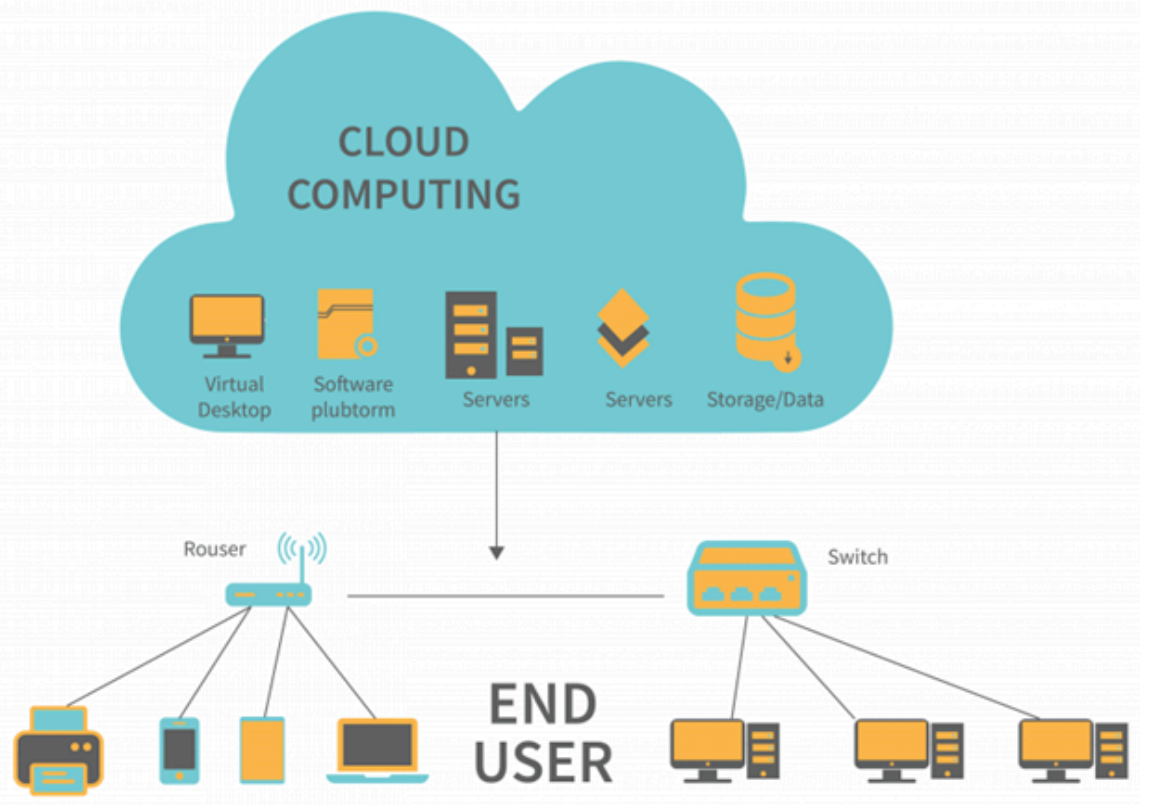
1. Infrastructure/platform-as-a-service (IaaS/PaaS): providing compute resources
2. Software-as-a-service (SaaS): all-in-one solution with online/API access
3. Cloud storage and data backup: allowing storage of (copies of) own data within centralized storage infrastructure



[Differences between IaaS, PaaS, and SaaS](#)

Data centers: activities

- Typical day-to-day use-cases:
 - Social media (e.g. Instagram)
 - Video streaming (e.g. Netflix, YouTube)
 - Webserver hosting (e.g. websites, live service video games)
 - AI training
 - Analyses and simulations (e.g. data analytics)
 - Software development and testing (e.g. GitHub)
- Practically anyone accessing the internet goes through a data center



Cloud is stored in the data center

Definitions: industry metrics

The Green Grid: industry consortium focusing on resource efficiency

- [Power usage efficiency](#) (PUE): ratio of total energy consumption, energy used for computing

$$\text{PUE} = \frac{\text{Total facility energy [kWh]}}{\text{IT equipment energy [kWh]}} > 1$$

- [Water usage efficiency](#) (WUE): ratio of water consumption, energy used for computing

$$\text{WUE} = \frac{\text{Annual water usage}}{\text{IT equipment energy}} \quad [L/kWh]$$

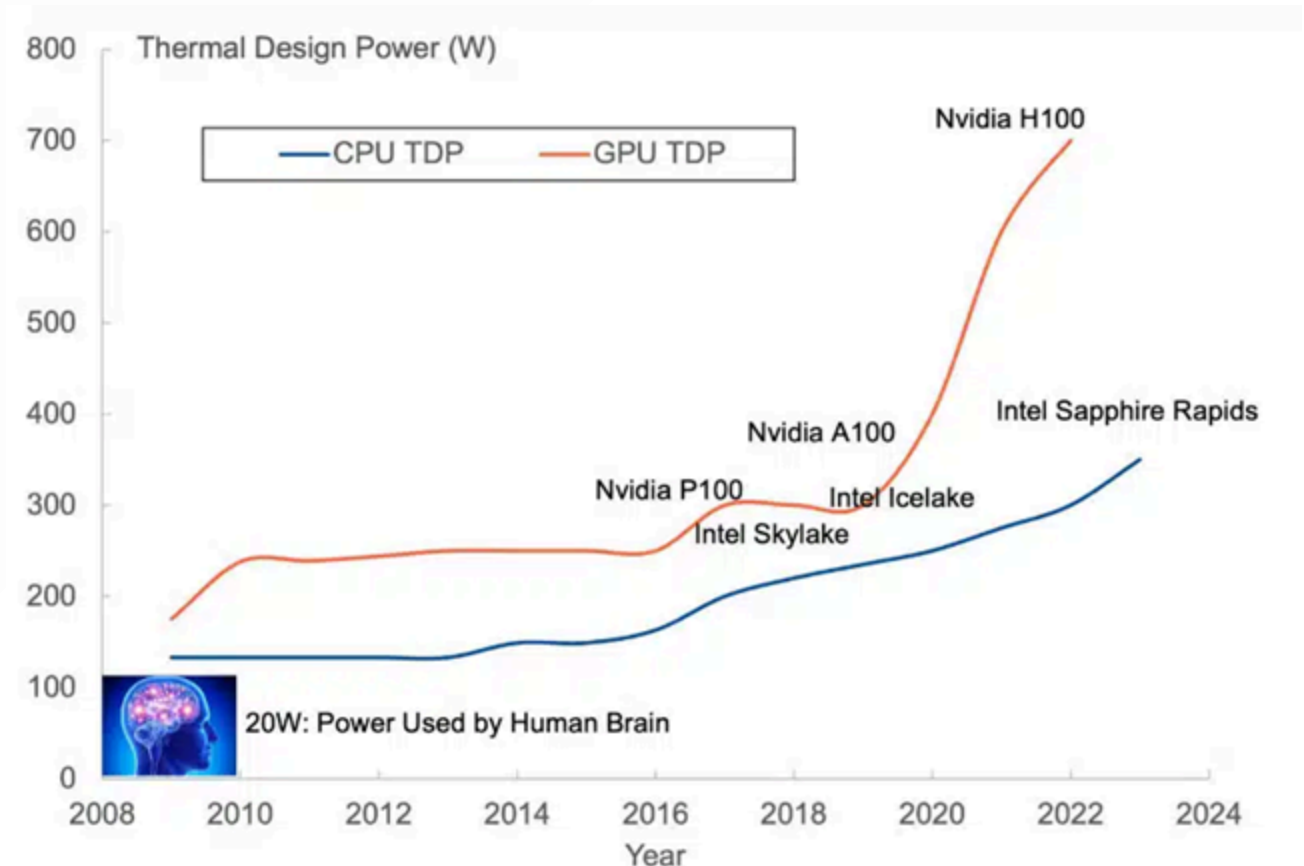
Uptime Institute: industry consortium focusing on data center reliability

- [Reliability](#): four tiers (I-IV); higher tiers more reliable; Tier IV: theoretical uptime of 99.995%

How does a data center work?

Data center architecture: computing

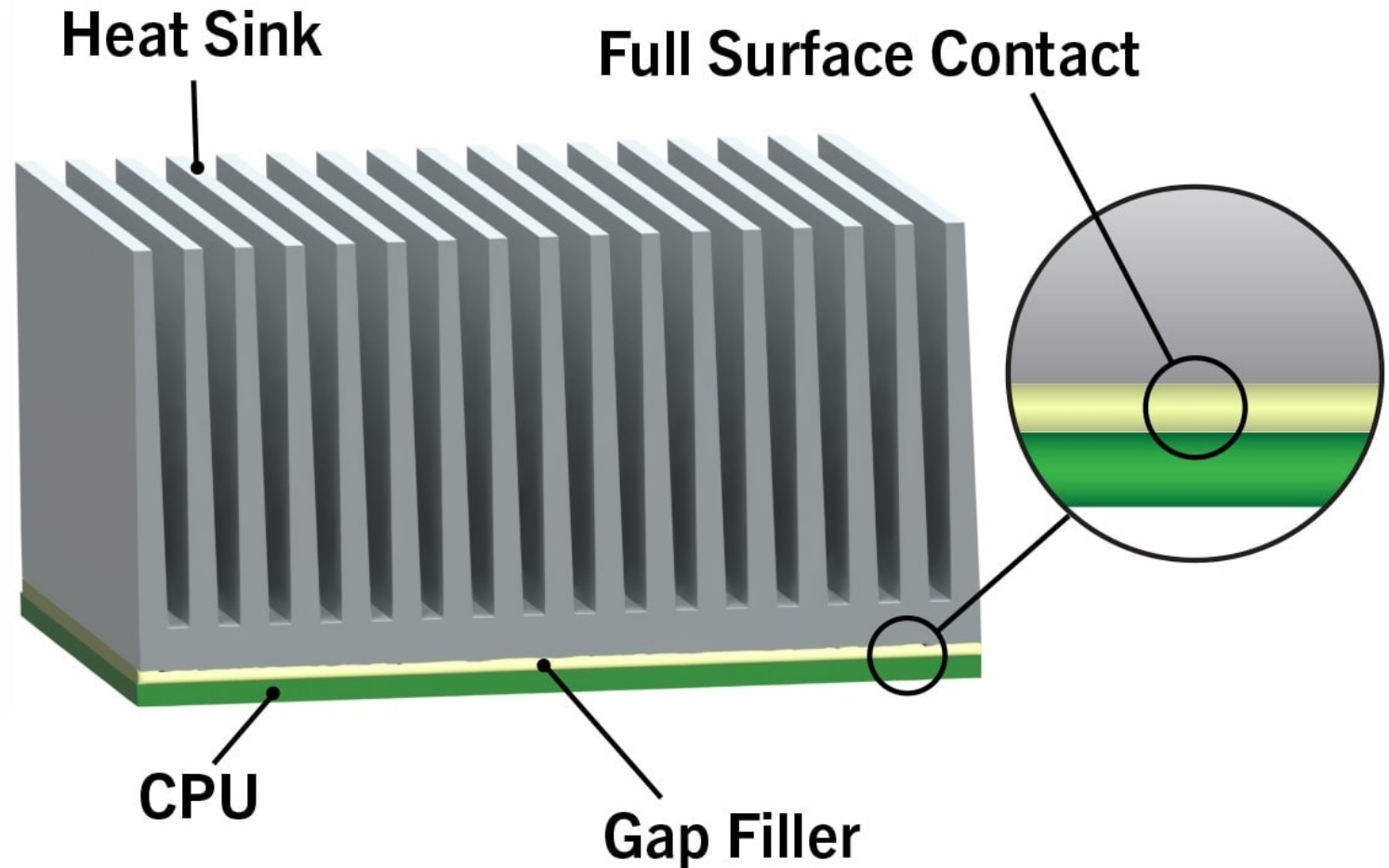
- Processor/chip: uses electrical power to perform computations, producing heat
- Chips designed for [thermal design power](#) (TDP): maximum allowable heat output
- Trend: increase in TDP, modern chips reaching [1500W](#)
- Computing activities depend on processor type (CPU/GPU/TPU) and workload type (web access, file streaming, AI training)



[Thermal design power ratings up to 2023](#)

Data center architecture: computing

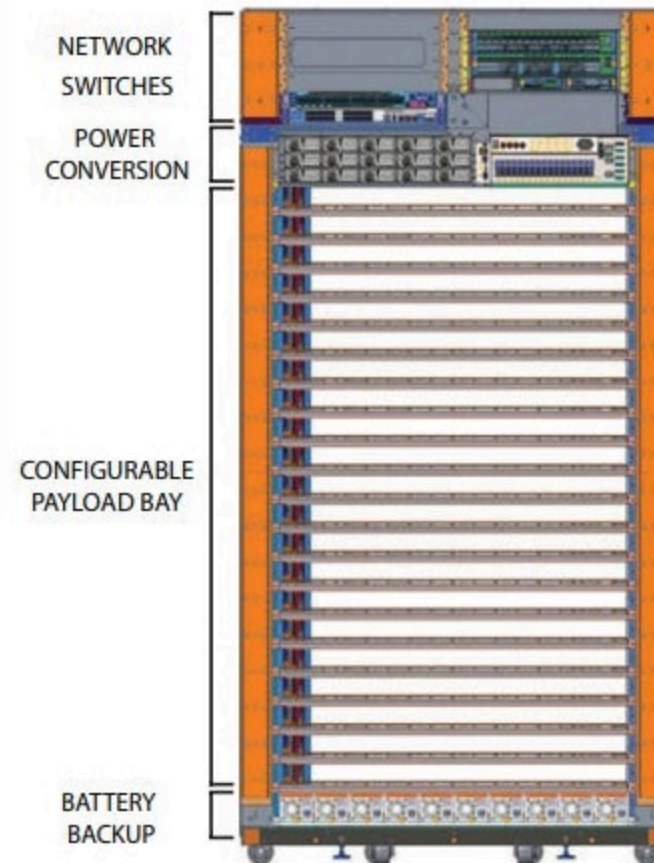
- Chip connected to thermal interface material (TIM), transfers heat to heat sink
- Heat sink has large surface area, radiates heat outward to surrounding air
- Hot air pushed out by server fan
- Higher TDP → [larger heat sink](#), more cooling required



[Thermal interface material and heat sink](#)

Data center architecture: computing

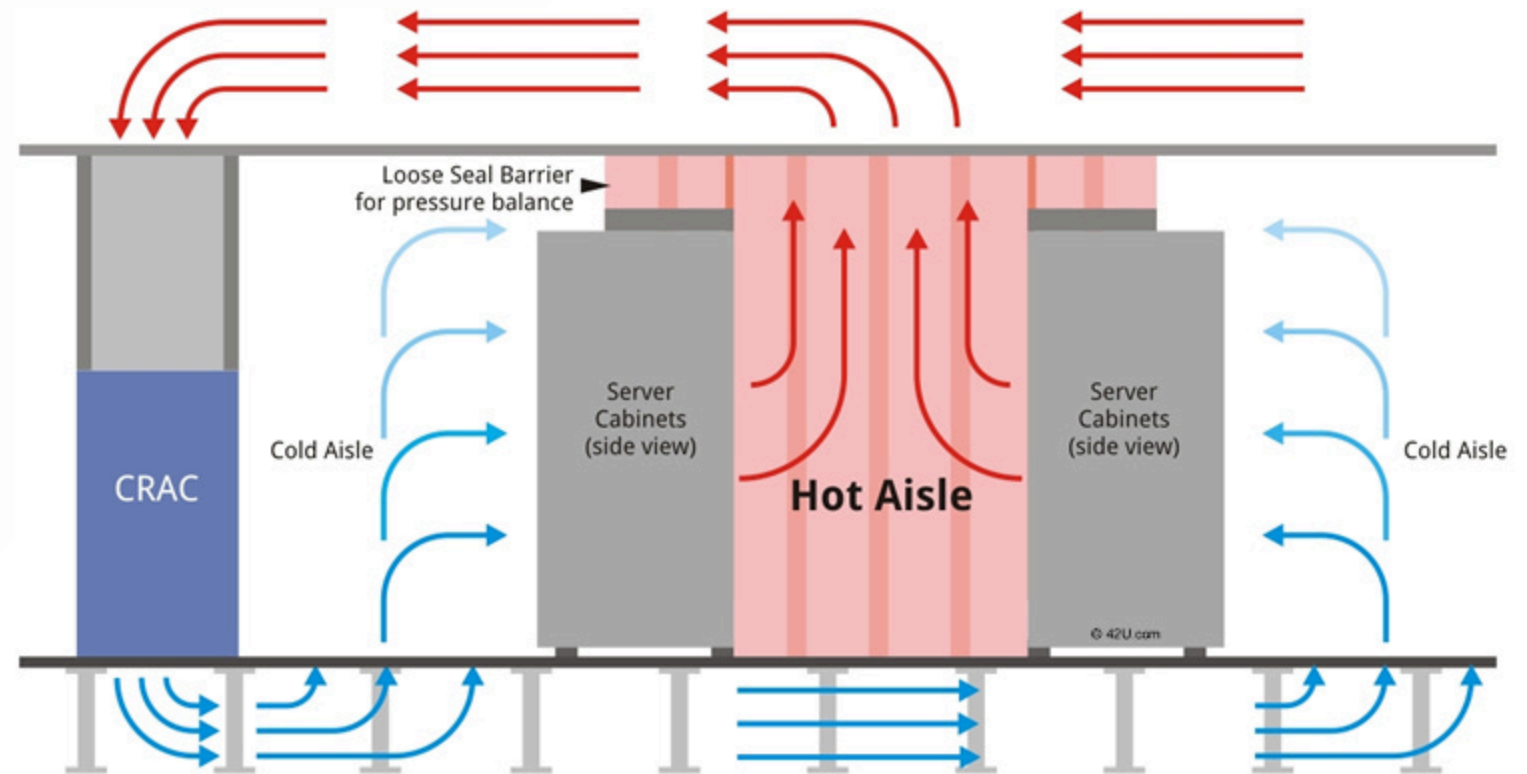
- Server/computer contains chips (CPU/GPU/TPU) and associated electronics, [networking](#), physical support, cooling fans
- Servers stored vertically in racks; several racks form a data hall
- Hyperscale data centers: up to [tens of thousands](#) of servers



[Server rack architecture](#)

Data center architecture: cooling

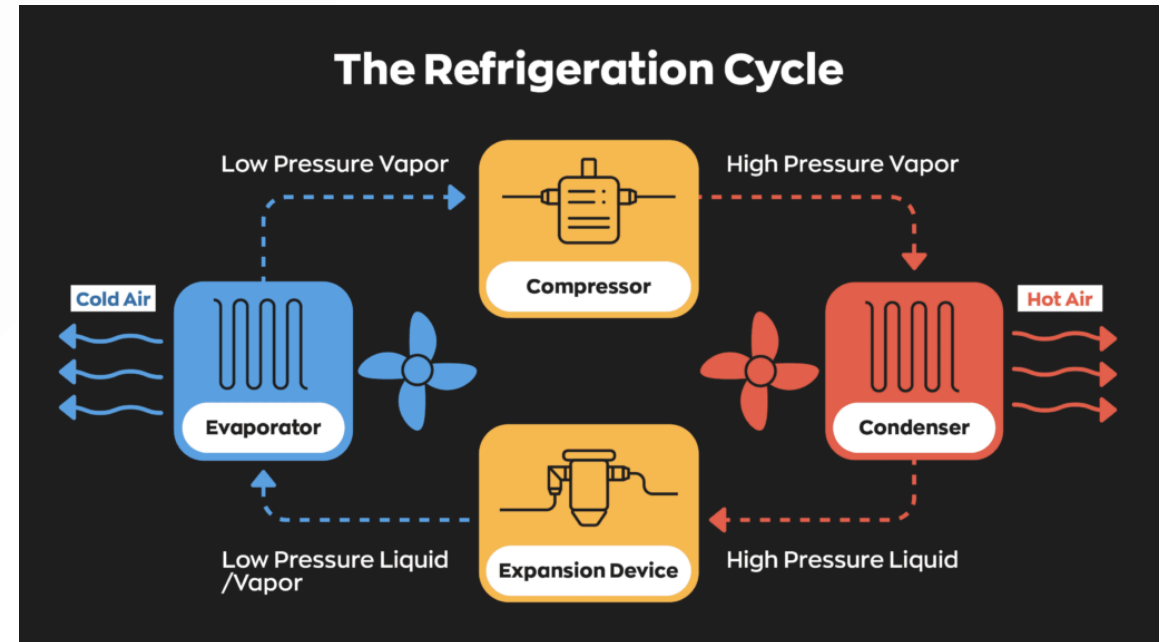
- Server fans evacuate heat to data hall
- [Computer room air conditioner/handler](#) (CRAC/CRAH) in each data hall blows hot air over cold water coil, transferring heat; cold air recirculated to data hall
- Modern data centers: fan walls, rear-door heat exchangers (RDHx) for each rack
- Avoid [mixing](#) between recirculated cold air and exhausted hot air → airflow management: aisle containment, computational fluid dynamics (CFD)



[Example: hot aisle containment](#)

Data center architecture: cooling

- Warm water from CRAC/CRAH transferred to chiller, cooled down, and recirculated
- [Chiller](#): large, energy-intensive, performs [refrigeration cycle](#):
 1. Liquid refrigerant absorbs heat from water, evaporates
 2. Refrigerant vapor increases in pressure, temperature in compressor
 3. Refrigerant vapor transfers heat to outdoor cooling loop in condenser
 4. Expansion valve reduces pressure, temperature



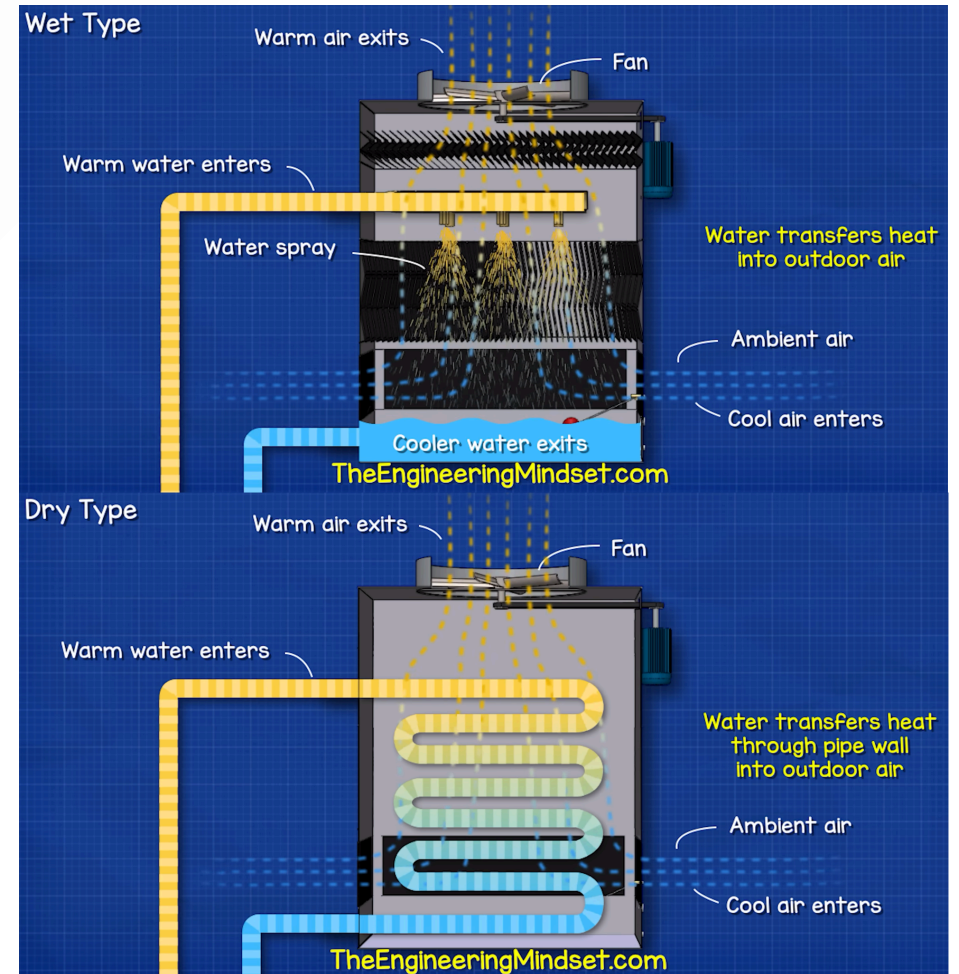
[Four-step refrigeration cycle](#)

Data center architecture: cooling

Heat moved from chiller to [external](#) cooling towers:

1. Wet cooling: water spray, evaporative cooling
 - High cooling capacity
 - Needs input water
 - Most effective: dry regions, low background humidity
2. Dry cooling: closed-loop, water flows through coil, cooled by fan
 - Lower cooling capacity
 - Consumes more electricity
 - Most effective: humid regions

Trade-off between energy and water efficiency!



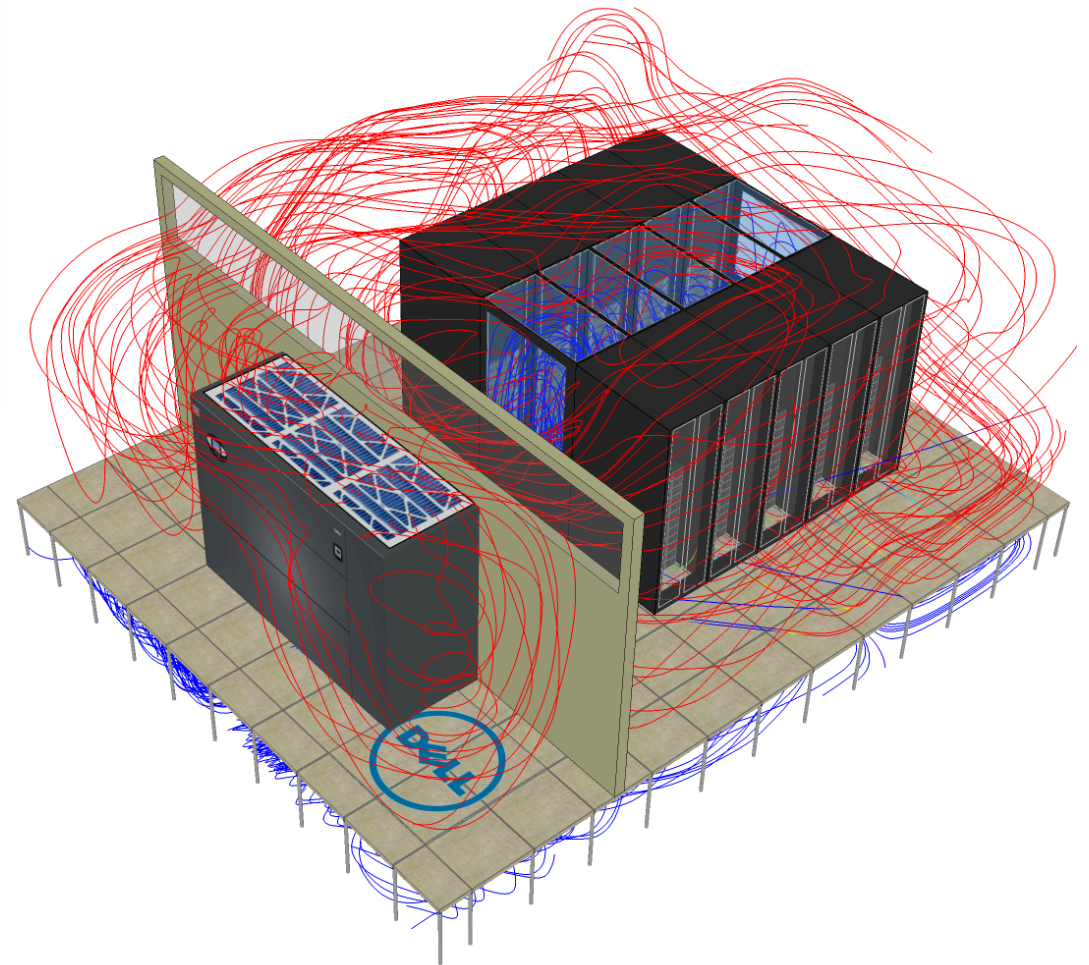
[Wet and dry cooling towers](#)

Data center architecture: cooling

Cooling energy efficiency gains: low hanging fruit [already_picked](#)

- Servers designed for up to 45 degree Celsius operation
- Larger fans: same airflow, less input power
- Optimized, contained airflow
- Free cooling: outside environment used to reduce need for cooling
- Multiple cooling loops, heat storage tanks

Takeaway: cooling system is complex and energy-intensive, but absolutely necessary; without it, [twenty minutes](#) until total heat death



[CFD modeling of data center airflows](#)

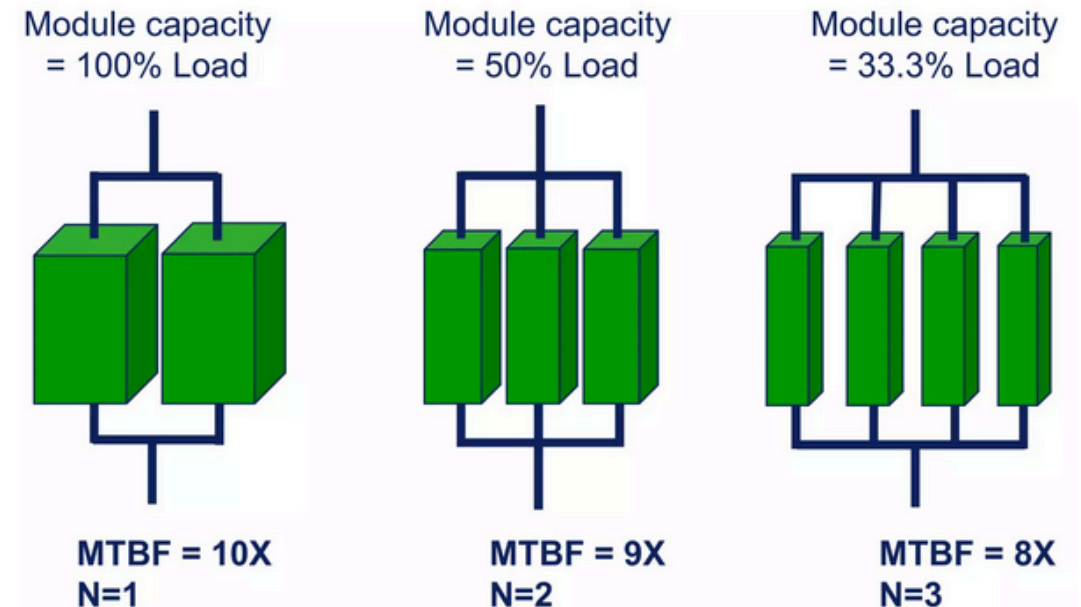
Data center architecture: power

- Sophisticated, redundant infrastructure delivers constant stable electricity to each server
- Power loss as function of current, voltage:

$$P = I [A] \cdot U [V] = I^2 [A^2] \cdot R [\Omega] \quad [W]$$

- Losses scale with current squared → [keep current low, voltage high](#) until data hall
- High voltage (HV) electricity from grid converted to medium voltage (MV) by power transformers
- Transformers in [redundant](#) N+1 configuration, partial operation (unused transformers deteriorate)

Redundancy: What is 'N+1'?

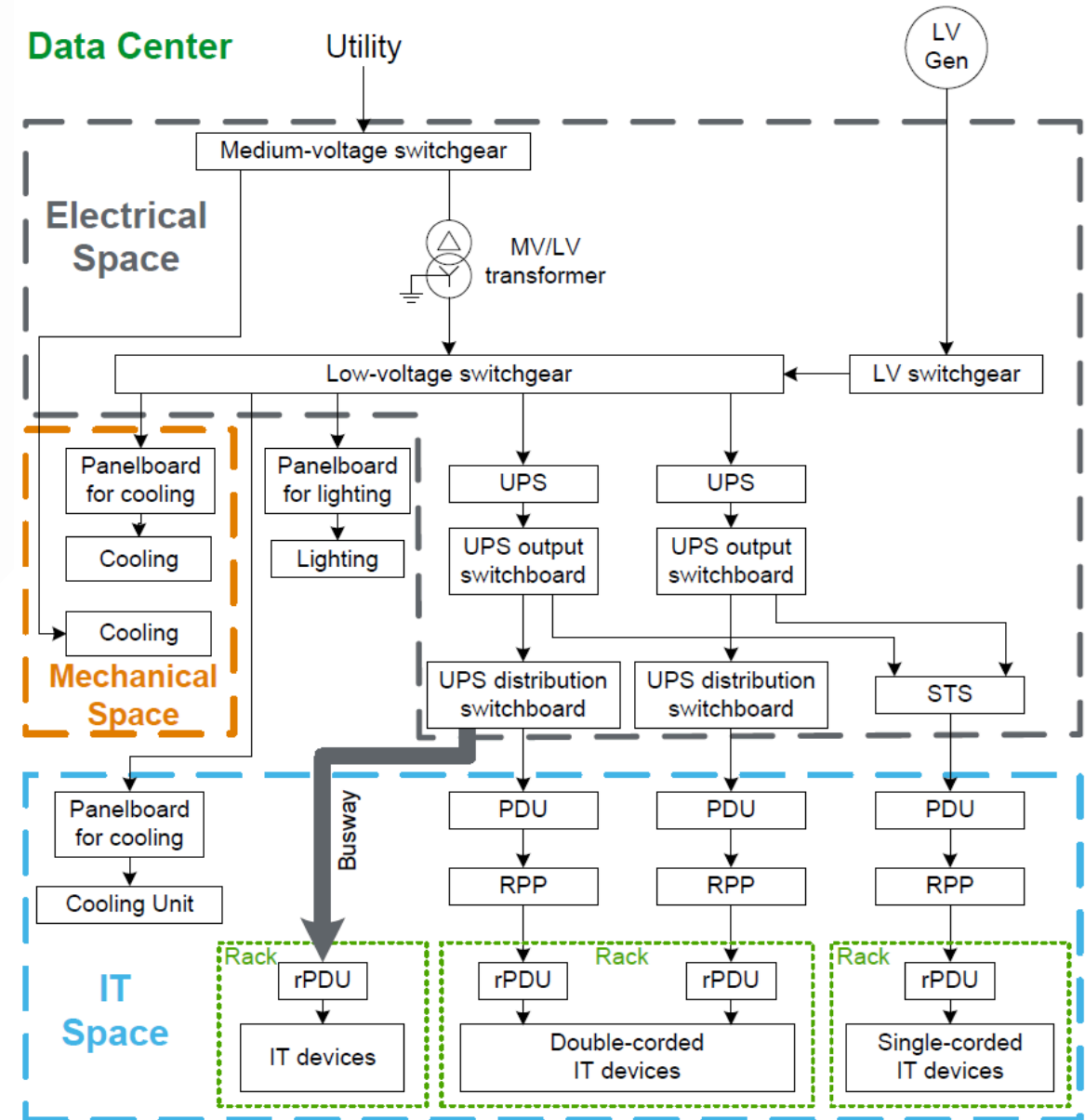


[Example: redundant transformer architecture](#)

Power diagram

Typical data center [power distribution architecture](#):

1. High-voltage (100kV) grid interconnection
2. Substation and transformers to medium-voltage (11kV)
3. Medium-voltage switchgear: also delivers power to cooling system
4. Transformer to low-voltage (415V)
5. Low-voltage backup diesel generator
6. Low-voltage switchgear: delivers power to lighting, uninterruptible power system (UPS) in each data hall



Data center architecture: power

- From transformers, [uninterruptible power supply](#) (UPS) to servers, and power to cooling system
- UPS distributes power to each data hall, data pods with dedicated switchgear, generator
- Switchgear distributes, protects, meters power
- Diesel generator with 24-48 hours of fuel for outages
- Downstream from switchgear: battery bank, takes over instantly during outages while generators start up
- Rack level: overhead busway/flexible power cables, two redundant power distribution units (PDU) to servers

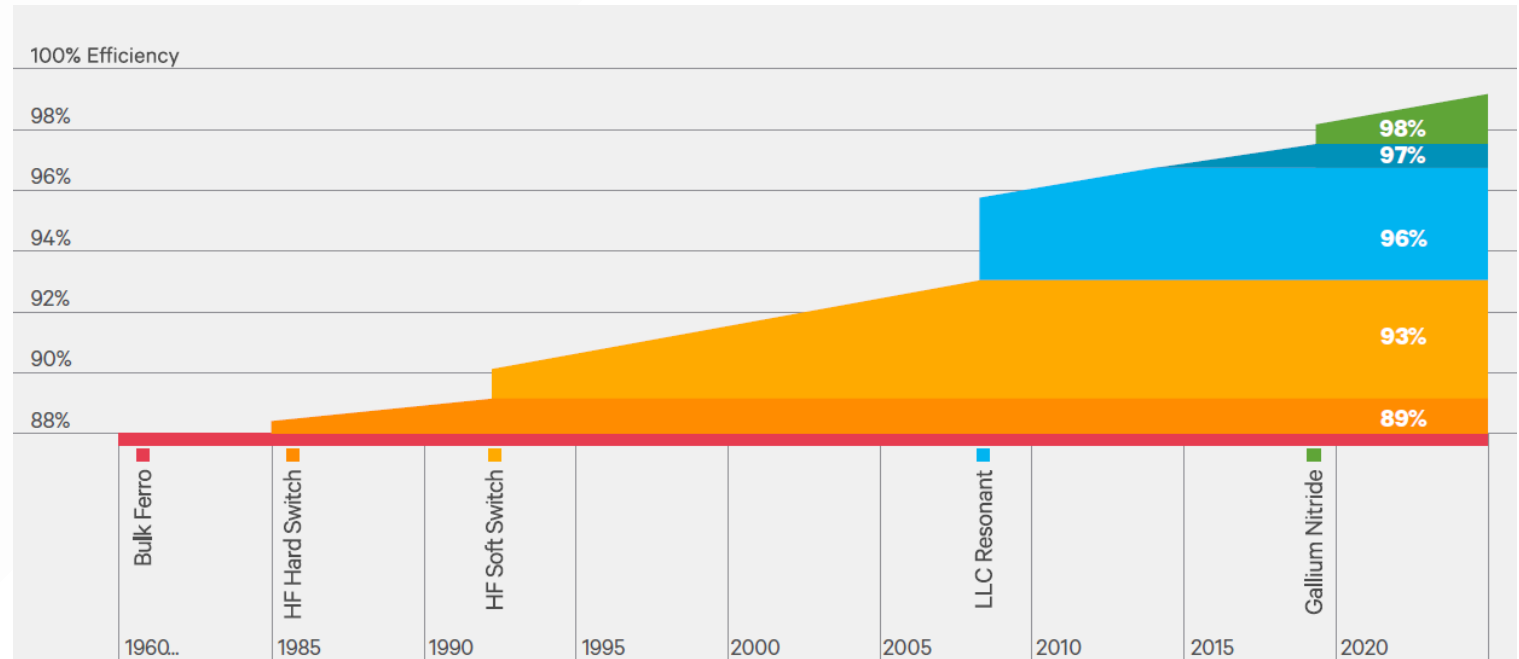


[Backup diesel generators in a data center](#)

Data center architecture: power

Power transmission efficiency gains: low hanging fruit already picked

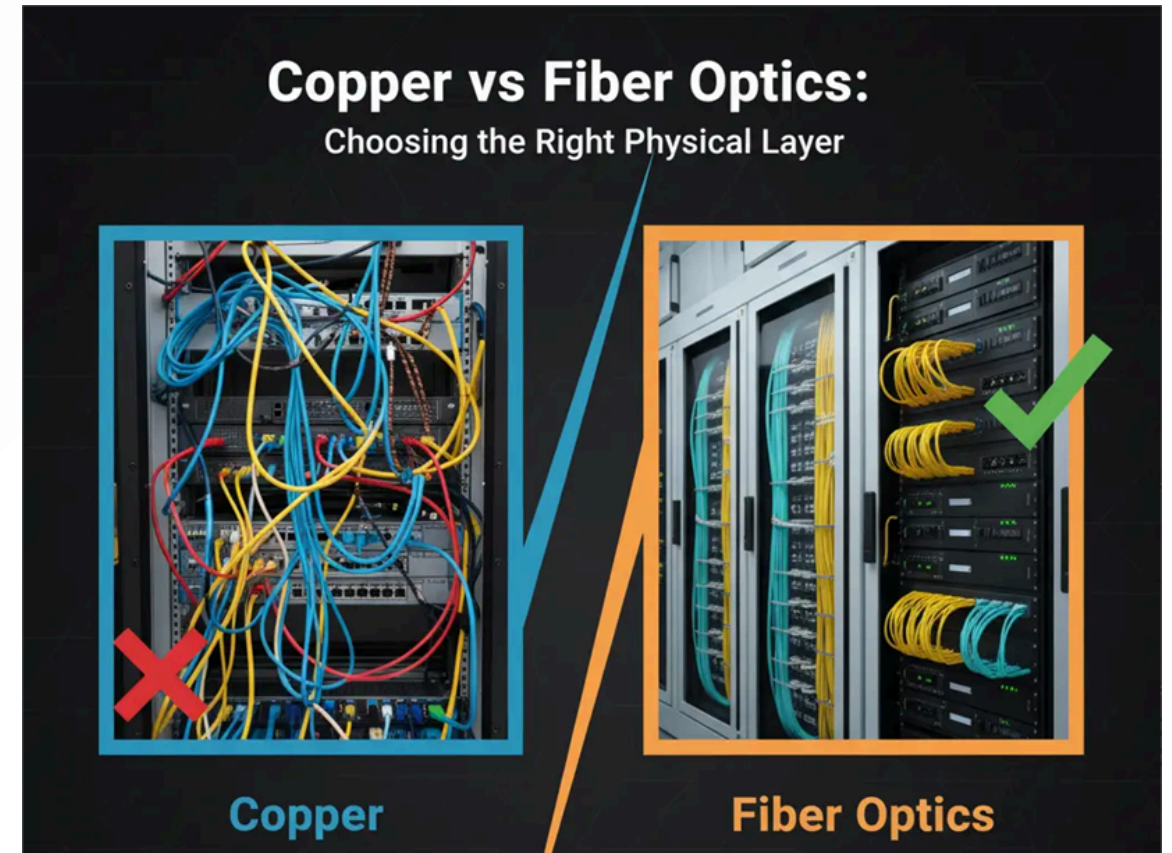
- Power equipment more efficient: AC-DC rectifiers [close to theoretical limit](#)
- Idle chips [ramp down](#)
- Smaller transistors have higher compute efficiency
- [Custom rack designs](#) reduce required battery capacity



[Efficiency trends for modern rectifiers](#)

Data center architecture: networking

- Networking used for data transfer within data center, but also to/from outside
- Different workloads require different networking architectures
- Trade-off between network performance and cooling: higher server density allows use of cheaper copper rather than fiber optic cables, but requires more cooling

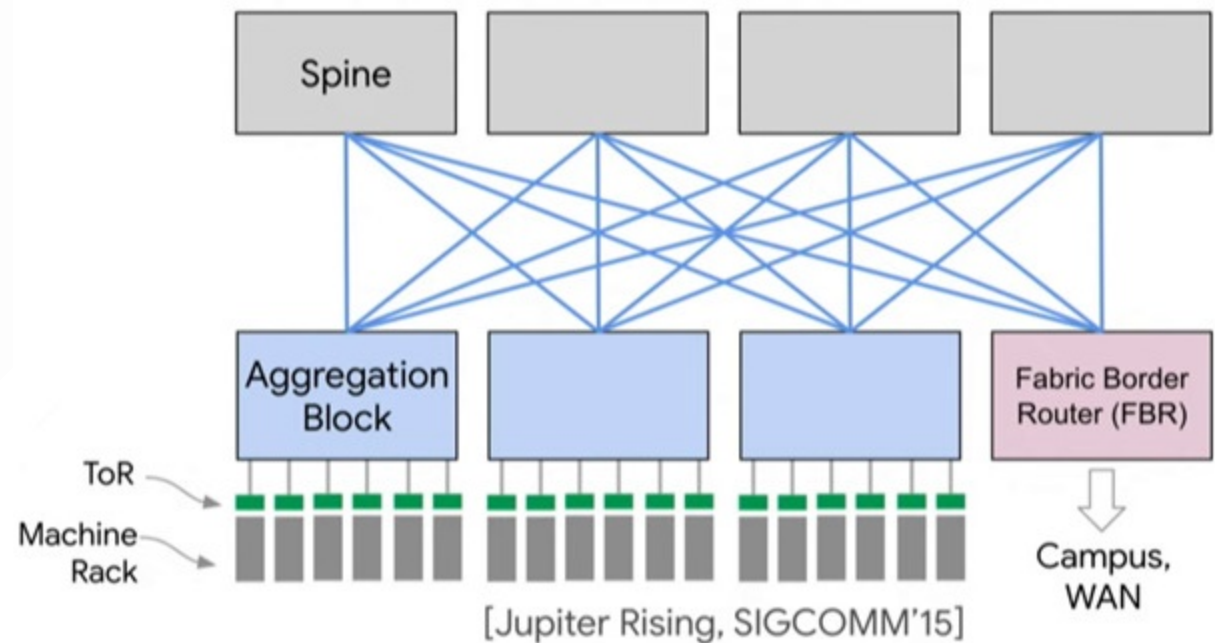


Big Fiber propaganda

Data center architecture: networking

Typical data centers: parallel workloads (video streaming, simulations, data storage)

- [Spine and leaf \(Clos\) topology](#):
 - Server rack connected to top-of-rack (ToR) switch
 - ToR switches feed into aggregation layers
 - Aggregation layers connected to spine/electronic packet switch (EPS)
- Continuous improvements in EPS performance result in doubling/tripling of power consumption

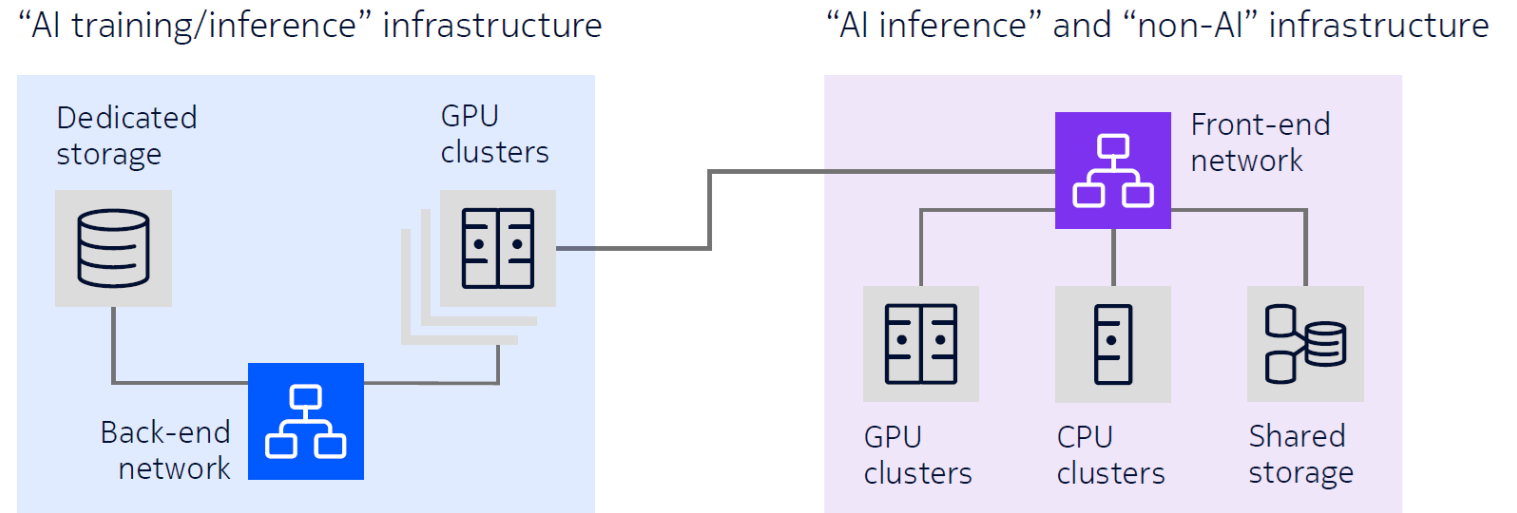


[Spine and leaf network topology](#)

Data center architecture: networking

AI training: GPU-GPU communication scaled out to thousands of racks

- Three [networking layers](#):
 - Frontend: internet connection, workload management (e.g. Slurm)
 - Backend: workload sharing between nodes
 - Out-of-band: server health, cooling, power control



[Backend networks are frequently used for training, while frontend networks are useful for inference](#)

- Most optimization effort targets backend topology, but also [frontend-backend integration](#)

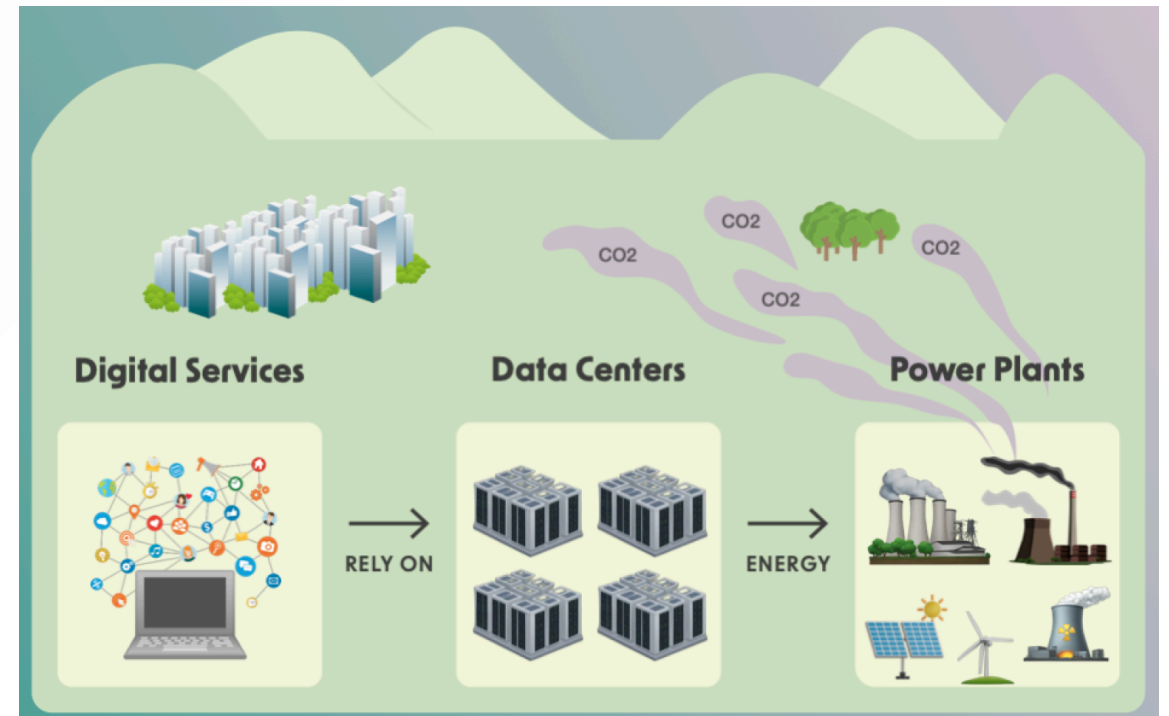
What are the impacts of the data center industry?

Data center impacts

- Data center industry has a significant energy/carbon/water/material footprint, affects local communities
- In this lecture: not just numbers, but also how to measure impacts
- Industry is [remarkably opaque](#) with e.g. power consumption → researchers rely on estimates
- Need to evaluate assumptions and methodologies critically!

Data center impacts: carbon footprint

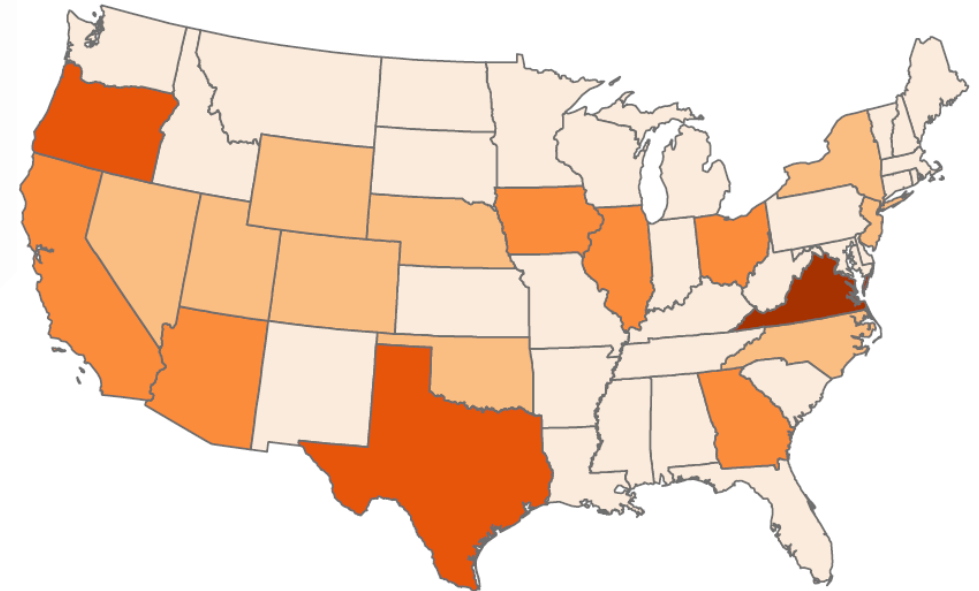
- [Alex de Vries](#): 3Wh (10.8kJ) per AI search-related inference
- [Shift Project](#): 3.7% of global GHG emissions caused by digital technologies
- [Factors](#) affecting data center carbon footprint:
 - Operational: electricity consumption
 - Operational: water consumption
 - Embedded: equipment lifetime



[Data center emissions result from their electricity consumption](#)

Data center impacts: carbon footprint

- Example: [large-scale assessment](#) (2024) of environmental impact of US data centers
- Methodology:
 1. Identified individual data center demand
 2. Found which power plants supplied each data center
 3. Estimated fuel mix for each power plant
 4. Attributed power plant emissions to each data center
 5. Made a neat [map](#)
- Result: data centers consumed >5% of 2022 US energy production



Data centers total CO₂e emissions (Megatonnes)

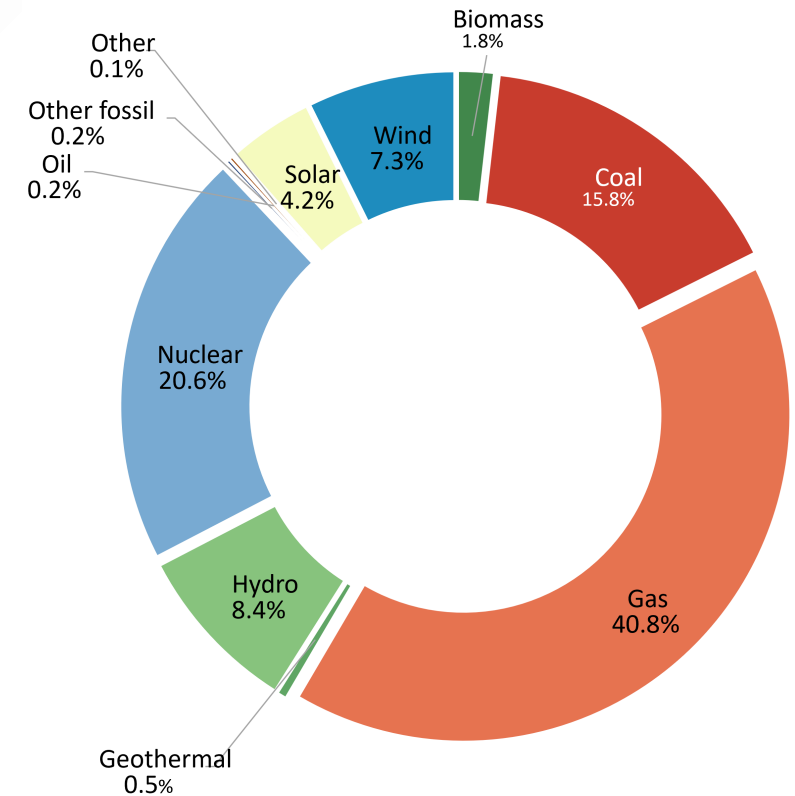


0.00 - 1.03
1.04 - 3.27
3.28 - 6.23
6.24 - 9.62
9.63 - 30.07

[US data center emissions by state](#)

Data center impacts: carbon footprint

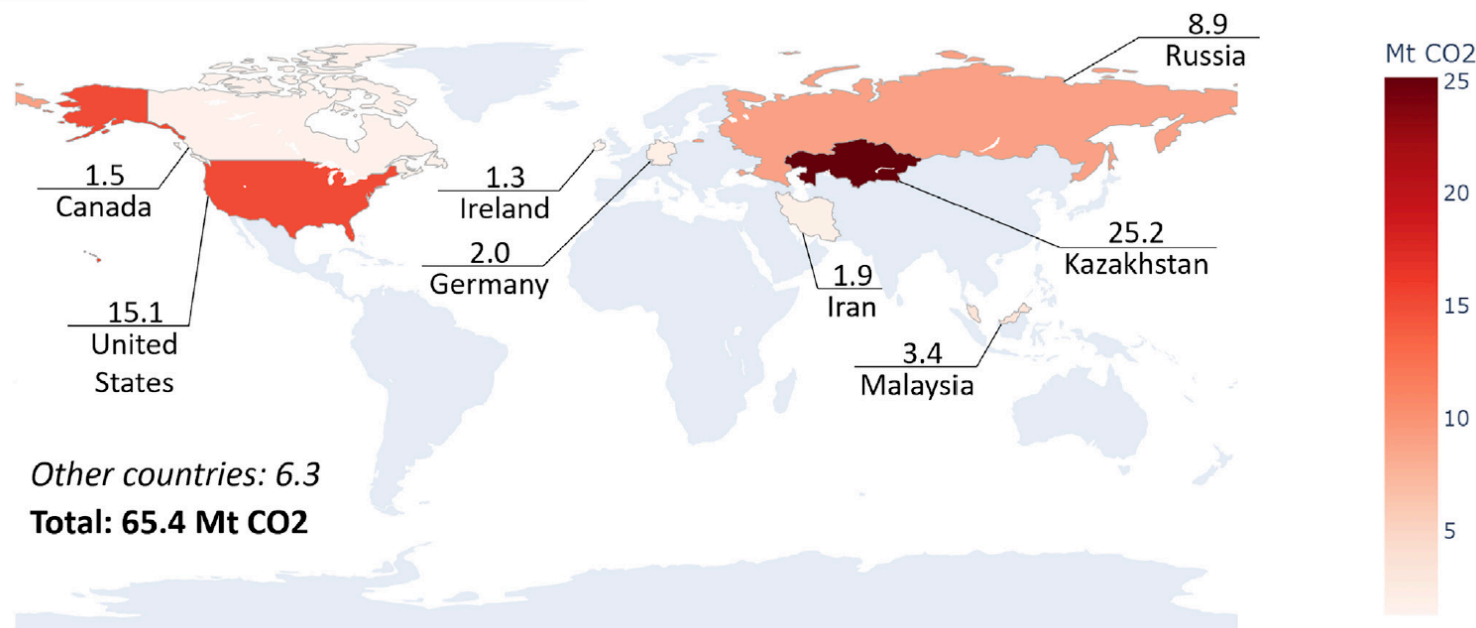
- Average carbon intensity: 548 gCO₂-eq/kWh, 50% higher than US average → data centers in fossil-heavy areas
- Fossil fuels account for >50% of energy consumed
- Attributional methodology: external data used, results attributed to each data center → [assumptions](#) made:
 - Assumed 0.75 utilization
 - No differentiation between colocation/ hyperscale
 - Power capacity estimated by size where data not available
 - Only operational carbon footprint calculated, bill of materials not available
 - Are these assumptions reasonable?



Estimated energy mix of US data centers

Data center impacts: carbon footprint

- [Similar methodology](#) (2025) applied to estimate carbon footprint of Bitcoin mining
- Miners reveal IP address → establish location, calculate carbon intensity of local electricity
- [Global study](#) (2021): carbon footprint of Bitcoin mining comparable to national emissions of Greece
- 2019: mining banned in China → miners migrated to Kazakhstan, US → more gas plants used



[Global Bitcoin mining carbon footprint in megatons](#)

Data center impacts: carbon footprint

- [Cloud Carbon Footprint](#) (CCF): open-source project for companies to calculate their cloud services carbon footprint
- Converts cloud provider usage data (compute, storage, networking) into energy consumption, estimates provider's data center PUE and carbon intensity:

1. $\text{CO}_2\text{eq}_{\text{total}} = \text{operational emissions} + \text{embedded emissions}$

2. $\text{operational emissions} = \text{cloud service usage } [use] \cdot \text{cloud energy conversion factors } [kWh/use] \cdot \text{cloud provider PUE } [-] \cdot \text{grid emissions factors } [metric\ tons\ CO_2eq]$

3. $\text{embedded emissions} = \text{estimated } \text{CO}_2\text{eq} \text{ emissions from server manufacturing}$

- Various estimates for [grid emissions factors](#), [manufacturing-related emissions](#) used → inexact results, but still useful

Data center impacts: carbon footprint

Example: edge AI inference system for wildlife observation

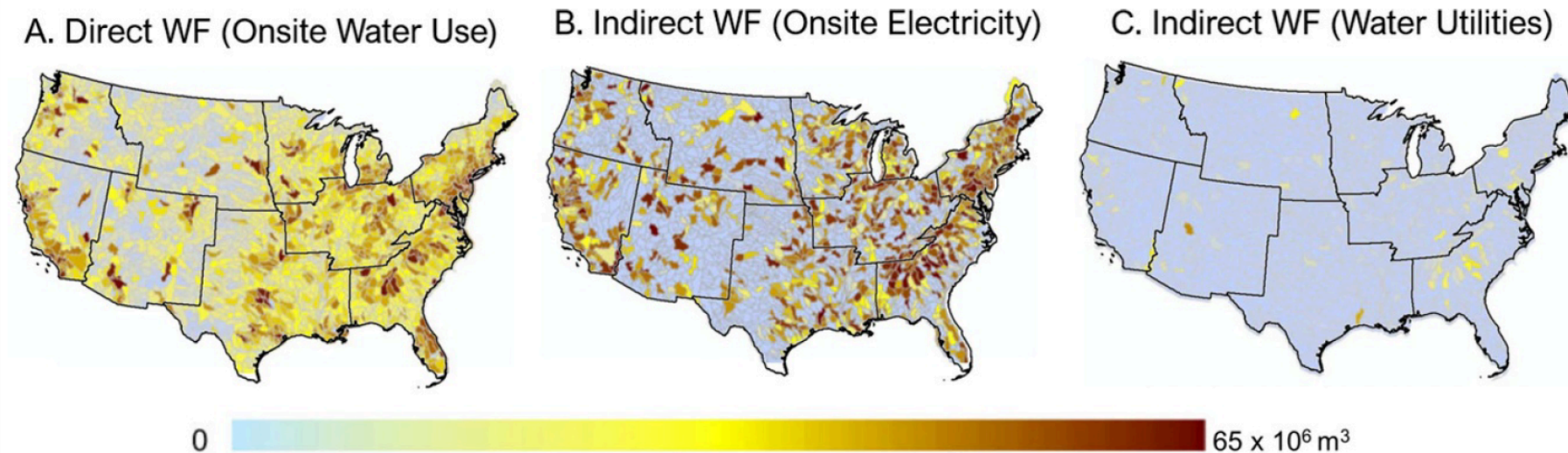
- All energy generated via own solar panels + batteries → no grid-related emissions!
- Embedded emissions: camera, processor, communications equipment, solar panel, battery (+transport to location, end-of-life...)



No grid connection = no grid emissions!

Data center impacts: water footprint

- [US estimates](#): data centers among top ten water-consuming activities, 20% draw water from already stressed sources
- [Latin America](#): significant recent buildout, including [drought-vulnerable](#) Mexico City
- Cloud providers do not report water consumption/local water stress levels → rely on estimates



[Direct and indirect 2018 water footprint of US data centers in cubic meters](#)

Data center impacts: water footprint

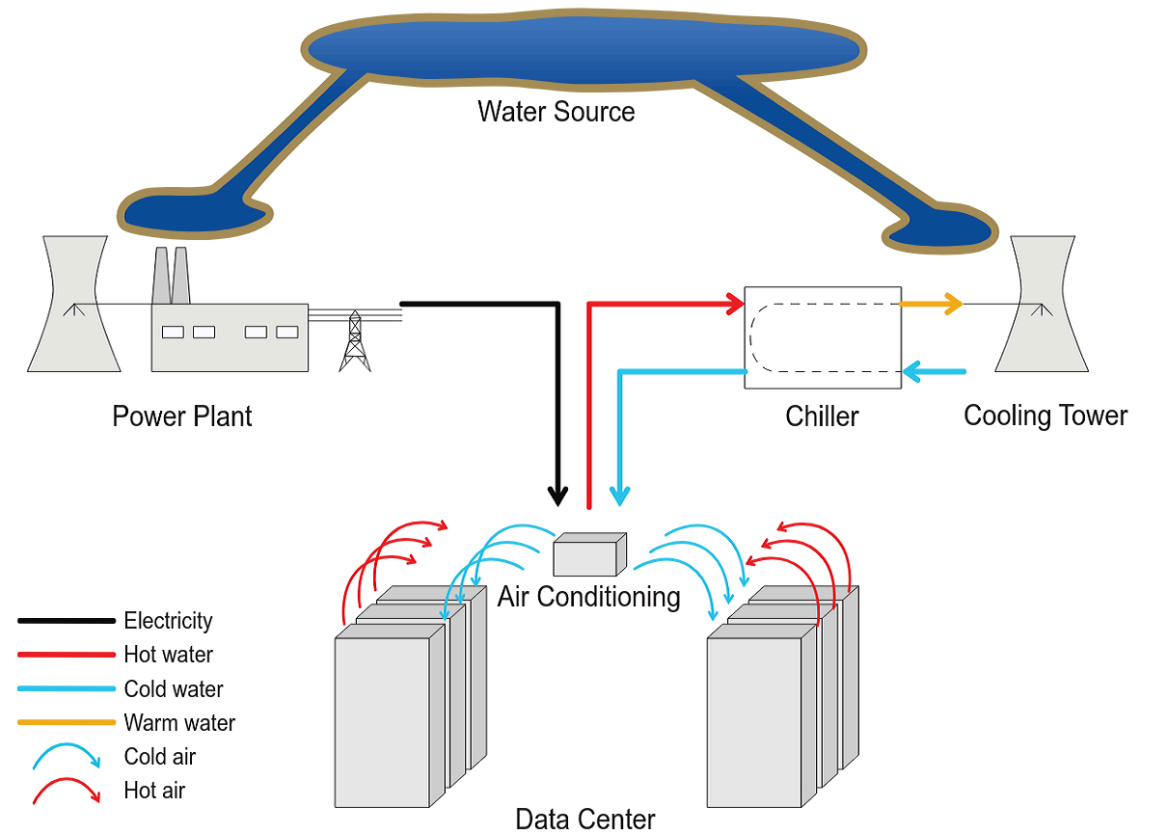
- [Methodology to estimate water consumption](#) of compute job M during time period t : sum of on-site water (for cooling) and off-site water (for energy generation):

$$W(M) = W_{\text{on}}(M) + W_{\text{off}}(M)$$

- On-site water consumption: function of energy used, data center WUE during job:

$$W_{\text{on}}(M) = \sum_{t=1}^T e(M, t) \cdot WUE_{\text{on}}(t)$$

- WUE estimated from external environment temperature: higher temperatures $\rightarrow$ more water needed for cooling



[Data center water flows](#)

Data center impacts: water footprint

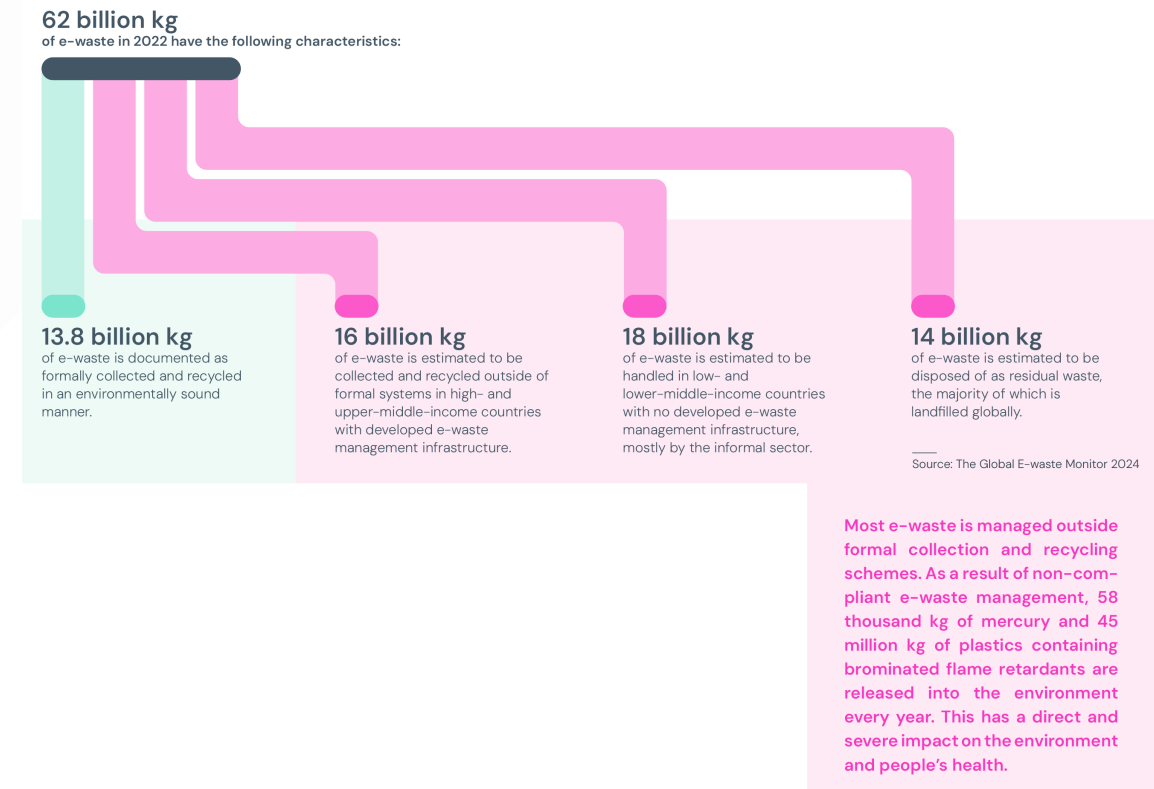
- Off-site water consumption: calculated as function of energy used, WUE (of power plant), and PUE (energy is also provided for cooling):

$$W_{\text{off}}(M) = \sum_{t=1}^T e(M, t) \cdot PUE(t) \cdot WUE_{\text{off}}(t)$$

- Power plant WUE: estimated based on fuel mix, relative water intensity factor (how much water required for 1kWh of energy)
- Takeaway: inherent operational trade-offs (e.g. using solar energy reduces CO₂ emissions, but works only during daytime, when temperatures are high, more cooling is needed)
- Some perspective: how does data center water use compare to agriculture? (hint: [it doesn't](#))

Data center impacts: material footprint

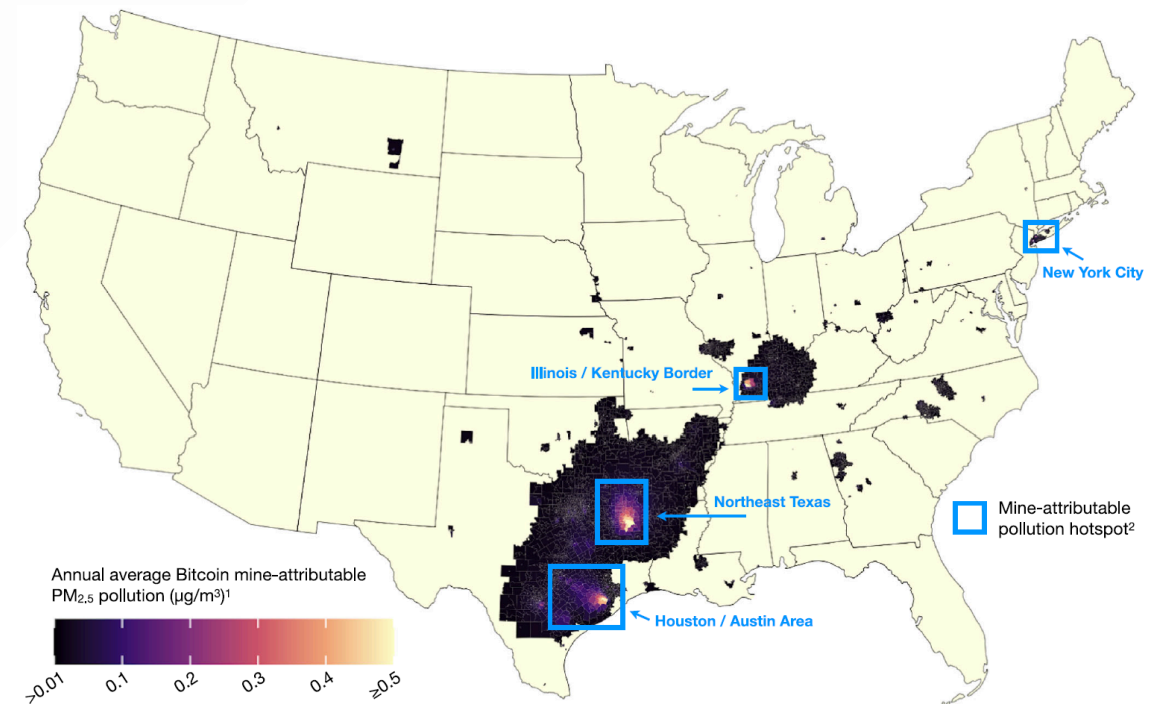
- Entire [frameworks](#) developed to calculate ICT equipment embedded emissions
- Lecture 6: broad range of substances in electronics → toxic byproducts from processing
- No public data about server lifetime (estimated [3-5 years](#)),
- Computing equipment usually not recycled → ends up in potentially toxic global waste stream (2022: [62 megatons](#))
- Also electrical, cooling systems, structural → up to [30%](#) of lifetime emissions embedded in physical infrastructure



[Global e-waste stream and recycled fraction](#)

Data center impacts: local communities

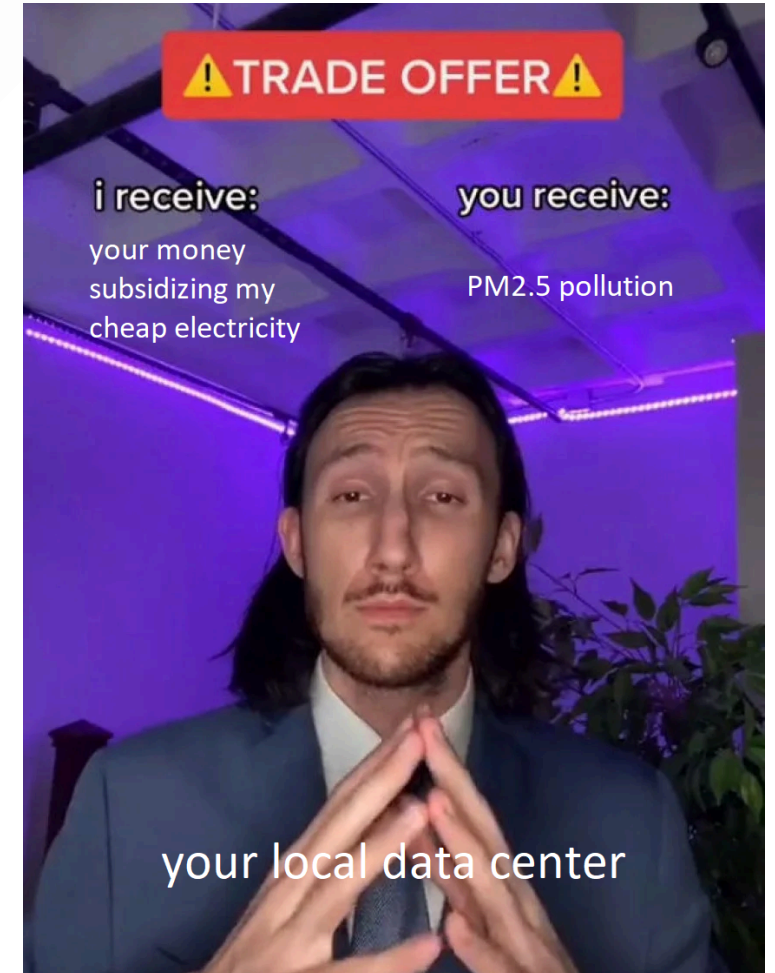
- Most energy powering data centers comes from fossil fuels
- Coal, gas plants produce fine air pollution e.g. [health-threatening](#) PM_{2.5}
- Bitcoin mining-attributed PM_{2.5} pollution: [46 million American citizens](#) in 27 states were found to be exposed to measurable concentrations in 2022-2023
- Unchecked water consumption for data center cooling can [deplete local aquifers](#) → decreased access for local agriculture, amenities, household activities



[Areas affected by PM_{2.5} pollution caused by bitcoin mining](#)

Data center impacts: local communities

- Utility companies [include costs of grid expansion](#) in residential/industrial electricity prices
- Data centers offer stable, long-term income → utilities offer [sweet deals](#) to data centers, shift infrastructure costs to regular consumers
- Data centers eat up power generation from local low-cost energy sources with [preferential agreements](#) → cost of electricity increases for average consumer (you)



Recap

- Data centers have become ubiquitous throughout the world, with practically all internet access going through them
- Complex computing, cooling, power, and networking architectures are required to keep data centers running
- Significant and increasing environmental impacts, estimates need to be made based on external data due to industry opacity
- Local impacts affect certain communities more than others

Exam(ple) questions

- 1. A 10MW data center has a yearly energy usage of 60GWh, of which 50GWh are used by actual computing activities, and consumes 250 million liters of water per year. What are its PUE and WUE?**
 - a. $PUE=1.2$, $WUE=5L/kWh$
 - b. $PUE=0.83$, $WUE=5L/kWh$
 - c. $PUE=1.2$, $WUE=4.17L/kWh$
 - d. $PUE =0.83$, $WUE=4.17L/kWh$
- 2. A data center has a PUE of 1.1, a utilization of 70%, and is rated for 20MW at peak load. If it is powered by a coal plant with a carbon intensity of $500gCO_2eq/kWh$, how many tons of CO2 can be attributed only to the computing activities in a year?**
 - a. 6.36 tons of CO2 per year
 - b. 7 tons of CO2 per year
 - c. 9.1 tons of CO2 per year
 - d. 10 tons of CO2 per year

Exam(ple) questions

1. A 10MW data center has a yearly energy usage of 60GWh, of which 50GWh are used by actual computing activities, and consumes 250 million liters of water per year. What are its PUE and WUE?
 - a. $PUE=1.2$, $WUE=5L/kWh$
 - ~~b. $PUE=0.83$, $WUE=5L/kWh$~~
 - ~~c. $PUE=1.2$, $WUE=4.17L/kWh$~~
 - ~~d. $PUE=0.83$, $WUE=4.17L/kWh$~~
2. A data center has a PUE of 1.1, a utilization of 70%, and is rated for 20MW at peak load. If it is powered by a coal plant with a carbon intensity of 500gCO₂eq/kWh, how many tons of CO₂ can be attributed only to the computing activities in a year?
 - a. 6.36 tons of CO₂ per year
 - ~~b. 7 tons of CO₂ per year~~
 - ~~c. 9.1 tons of CO₂ per year~~
 - ~~d. 10 tons of CO₂ per year~~

Feedback

- We need your feedback to continue improving the course!
- Feedback on Lecture 5.1 slides/notes:



Thank you for your attention!

Next lecture: Sustainable Cloud Computing, Part 2

Thursday, October 1, 15:45-17:30, EEMCS Hall F

- Industry trends: compute growth, efficiency gains, energy
- Prospective solutions: cloud strategy, demand response, cooling technologies, waste heat reuse

Questions?